LifeLink

A Secure Infrastructure-Independent Emergency Communication System

D2 Project Specifications and Literature Review

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1 Introduction

1.1 Description

LifeLink is an emergency communication prototype for situations in which mobile networks and internet access are unavailable. Earthquake rescue is the primary design and evaluation scenario. We aim to connect people who can actively request help and people who may be unable to use a phone or press a button.

The system combines a low-power wristband, a separate offline messaging channel, and a local gateway with a rescue-team dashboard. The wristband attempts an optical pulse measurement and reports whether the estimate is usable or uncertain. A manual help request remains independent of measurement success. The messaging channel supports a short request and an operator reply.

Communication depends on powered local equipment and a usable radio path. A gateway receipt and an acknowledgement issued by a rescue operator are distinct events. Missing messages or uncertain measurements never establish that no living person is present. The prototype supports independent search and rescue; it does not diagnose a person’s condition or assign rescue priority.

Project objective

We will evaluate how help messaging, pulse-quality assessment and bounded traffic control work together under limited energy and airtime. The requirements below define the intended behaviour and how it will be checked. Component and radio choices remain subject to feasibility work.

1.2 System scope

Component	Planned responsibility
Wristband	Manual help input, rule-based motion activation, periodic PPG and acceleration acquisition, quality-labelled reports, device-presence response and vibration feedback. A removable capsule and soft interchangeable strap are the initial form concept.
Offline messaging channel	Short help messages and operator replies. A gateway-hosted Wi-Fi browser page and a radio-based interface remain alternatives. An internet-disabled messaging flow is part of the first-semester demonstration.
Gateway and dashboard	Authenticate wearable traffic, display measurement age and communication state, manage message queues and time-limited restrictions, and relay an actual operator acknowledgement.
Field relay nodes	A possible second-semester extension for collection and forwarding, subject to radio, budget and integration results. They are not required for the first direct-to-gateway demonstration.

The wearable–gateway link and the separate messaging interface are distinct choices. LoRa is the preferred wearable-link candidate; Wi-Fi/BLE is a first-demonstration fallback. A fallback demonstration checks application behaviour, not LoRa range, airtime or energy performance. Transfer to LoRa would require separate integration and verification.

Semester boundaries

Semester one: requirements, analysis and high-level design; one wristband, one messaging client and one gateway; rule-based quality assessment; authenticated temporary identifiers; a timed restriction of routine reports; operator acknowledgement; and an initial quality model evaluated on a computer. The initial demonstration includes both the wearable flow and the offline messaging flow.

Semester two: deploy the feasible quality approach on the selected MCU; measure processing time, memory and energy; expand range, obstruction and multi-device traffic experiments; and investigate automatic restriction and field relays where feasible. The model and a larger network are not claimed to be working first-semester outputs.

Scope limits

Precise indoor positioning, continuous tracking, respiration radar, room imaging, medical diagnosis and rescue-priority decisions are outside the initial scope. The first prototype requires no personal name or account. Optional personal or medical profiles are a future extension, not a first-prototype requirement. The wearable remains external and removable.

Operational states

The dashboard distinguishes pulse indication detected; device detected with uncertain measurement; and a missing expected response from a previously observed device. Reporting restrictions are a separate communication status. While a confirmed restriction is active, the dashboard shows the planned pause and an approximate return time rather than interpreting the planned silence as an unexplained missing response.

A displayed pulse indication is an accepted sensor estimate with a timestamp. It is not a medical determination. No response from a known device is different from finding no devices in an area; neither is used to conclude that an area contains no survivors.

1.3 Constraints

Constraint	Design consequence
Economic	Funding, laboratory inventory and borrowing arrangements are not confirmed. The earlier 7,000–10,000 TL range is a provisional planning allowance, not a quotation or agreed team budget. Buy in stages; check taxes, availability and compatibility; borrow reference and power-measurement instruments where possible.
Environmental	Dust, ambient light, temperature, moisture and mechanical loading may affect operation. Record test conditions and observed failures. No waterproof, rubble-proof or certified environmental rating is claimed for an untested prototype.
Social	Daily and overnight wear must be acceptable. Compare strap fit, pressure, skin comfort and button access through the applicable participant process. Use an accessory-like design. People without a device or usable connection remain within independent rescue searches.
Political and regulatory	The communication path must remain useful when public networks or online platforms are unavailable. Select radio settings against the applicable Turkish BTK criteria, including power, channel access and any duty-cycle limits. Budget device transmissions and gateway downlinks separately [11].
Ethical	Incorrect certainty, stale measurements and false acknowledgements can mislead users. Explicit uncertainty, freshness labels and operator-issued acknowledgements are required. Traffic controls must preserve a bounded opportunity for manual help.
Health and safety	Protect against battery short circuit, heating and mechanical damage. Use compatible power and charging components if a rechargeable cell is selected. Human measurements and comfort studies follow the applicable TEDU ethics process [12]. A demonstration is not a medical-device validation.
Manufacturability	Begin with documented modules, secure wiring and a simple enclosure. Check sensor identity, optical isolation, antenna placement and skin contact. Development-board size and standby consumption are reported as prototype properties, not final wearable performance.
Sustainability	Aim for serviceable electronics, replaceable straps and an appropriate battery-maintenance plan. Long standby life depends on the cell, self-discharge, temperature and ageing as well as circuit current. Firmware, configuration and build documentation must remain maintainable.

Airtime illustration only: if the selected sub-band has a 1% duty-cycle limit, a 1 s transmission every 60 s gives 1.67%, whereas every 300 s gives 0.33%. Neither example establishes the actual permitted configuration. Payload, spreading factor, retries and downlinks must be included in the measured budget; gateway acknowledgements consume the gateway’s own airtime.

1.4 Professional and ethical responsibilities

Honest reporting and prevention of harm

The ACM Code of Ethics, IEEE Code of Ethics and joint Software Engineering Code place public welfare, honesty, privacy and professional responsibility at the centre of engineering work [1–3]. We apply these principles through explicit uncertain states, visible measurement age and a clear distinction between a local action, gateway receipt and operator acknowledgement. Test results will include failures and operating conditions; laboratory delivery does not demonstrate reliable communication through rubble.

Privacy and security

Wearable reports do not require personal identity. Temporary identifiers are authenticated, but the authorized gateway may associate reports with the same device; this is pseudonymous operation rather than a guarantee of complete anonymity. Pulse payloads require confidentiality as well as authentication. Keys and sensitive participant records must not appear in the public repository. Access, retention and deletion rules for study data will be documented before collection. Applicable data-protection requirements, including health-data handling, must be reviewed [13].

If personal profiles are developed later, voluntary field-by-field sharing, revocation and authenticated recipients are product principles. Consent is not presented as the only possible legal basis for every emergency use. Personal profiles and continuous location tracking remain outside the initial demonstrator.

Participants and inclusive evaluation

We will confirm the university's ethics requirements before live human measurements or comfort evaluation. Required approval and informed consent must precede the relevant activity. Until then, development can use licensed public data, clearly labelled prerecorded signals and hardware tests without participants. Results from a small study or daily-activity data will not be generalized to all skin tones, tattoos, injuries, cold conditions or disaster victims.

Attribution and responsibility

Cite research, datasets and reused software, comply with their licences, and record team contributions. Use restricted records for participant data and public documentation for submitted reports and non-sensitive implementation material. Claims about novelty must follow the reviewed evidence.

Open implementation decisions

Decision	Basis for selection
Messaging interface	Compare a local browser page with a separate radio interface for access, two-way use, hardware cost and effort.
Wearable–gateway link	LoRa is the preferred candidate, not a confirmed purchase. Wi-Fi or BLE may support the first functional demonstration if sourcing or setup delays LoRa. Select by availability, cost, bidirectional operation and development effort.
MCU and sensors	Laboratory stock is not yet confirmed. Check documented drivers, dimensions and complete-board current. MAX30101 and ADXL362/LIS2DH12 remain candidates.
Power and enclosure	Use USB power for initial bench integration. Select a battery for the portable demonstration after measuring consumption and checking dimensions, pulse-current capability and protection. Avoiding routine recharging remains a longer-term goal to evaluate.

2 Requirements

These are initial specifications. S1 and S2 identify planned implementation stages, not completed work. Demonstration checks are bounded tests. Timing, message-size, current and quality thresholds still requiring experiments will be selected and recorded in D4; they are not guaranteed performance values.

2.1 Wearable and information states

ID	Stage	Requirement	Verification
FR-01	S1	The wristband shall activate through the help button or a rule-based motion trigger. The first demonstration shall use a configured long press and a controlled motion input.	Apply each trigger and record the transition from standby.
FR-02	S1	A manual help request and a device-presence report shall not depend on a successful pulse measurement.	Supply invalid or absent PPG input and trigger help; observe the report.
FR-03	S1	In emergency mode, the device shall acquire timestamped PPG and acceleration windows at configured intervals.	Inspect sample counts, timestamps and acquisition gaps.
FR-04	S1	The rule baseline shall label each evaluated window usable or uncertain, recording observable reasons. A failed estimate shall not be reported as a valid pulse value.	Replay usable, flat, saturated and motion-contaminated inputs.
FR-05	S1	The dashboard shall distinguish accepted pulse indication, detected device with uncertain measurement, and missing expected response from a known device.	Inject each condition and check the displayed state.
FR-06	S1	Measurement age and last-contact time shall be displayed separately; old measurements shall be marked stale under a configured freshness policy.	Continue contact without new measurements, then stop contact.
FR-07	S1	The system shall distinguish locally recorded request, gateway receipt and operator acknowledgement. The wristband shall indicate an operator acknowledgement only after receiving its authenticated message.	Withhold the operator action or its return packet; verify that no operator confirmation appears.
FR-08	S1	Vibration shall be the main feedback channel, with an LED as a secondary channel. Local request and operator acknowledgement shall use distinct configured patterns.	Exercise the documented patterns and inspect the command log.

Reason codes describe observable signal conditions, such as flat input, saturation, high movement or weak spectral concentration. They do not diagnose absent circulation, arrhythmia or sensor detachment from one signal feature alone.

2.2 Authentication and bounded traffic control

ID	Stage	Requirement	Verification
FR-09	S1	Wearable reports and control messages shall be authenticated. Reports shall carry a temporary device identifier, message counter, message type and measurement timing information.	Inspect packet fields and verify valid test messages.
FR-10	S1	The receiver shall reject failed authentication, altered packets and stale unauthorized replays. Valid duplicates of the last applied command shall follow FR-14.	Use a wrong authentication tag and an older command counter; check rejection logs.
FR-11	S1	An authorized operator shall be able to request a bounded pause of routine automatic reports. The command shall specify a duration and expire at the device; routine reporting shall resume automatically.	Issue a short pause and observe application, expiry and resumed reporting.
FR-12	S1	A routine-report pause shall not disable manual help. Manual calls shall have a separate small rate budget and remain subject to legal airtime and link availability.	Press help during the pause; verify independent eligibility and rate limiting.
FR-13	S1	During the pause the device shall retain sparse independent control-listening opportunities. The gateway shall queue commands for available windows and expire undelivered commands at a configured deadline.	Disable routine reports; exercise an independent window without a button press.
FR-14	S1	An authenticated duplicate of the last applied command shall not reapply it or restart its timer. The device shall return its status and remaining duration again. Older or altered commands shall not change state.	Drop the first application acknowledgement and resend the same command in FR-13’s independent window.
FR-15	S1	The dashboard shall distinguish queued command, confirmed receipt if separately reported, and confirmed application. It shall mark restriction as active only after application confirmation.	Suppress confirmation; verify an unconfirmed state rather than active restriction.
FR-16	S1	While a confirmed pause is active, show restriction status, last measurement age and an approximate return time. The return estimate shall use the device-reported remaining duration and gateway receipt time.	Verify planned silence and stale measurement display during the pause.

A receipt-only response is optional in the initial implementation. If the device returns only an application response, the dashboard shall not invent an intermediate delivery confirmation. Return times remain approximate because communication delay and clock differences are not eliminated.

2.3 Further traffic and messaging requirements

ID	Stage	Requirement	Verification
FR-17	S2	A newer valid restriction command shall replace the previous one and start a new bounded timer at application. An authenticated early-release command shall follow the same queue and replay rules; release after expiry shall return an already-ended status.	Test replacement, early release and release after expiry. These are designed extensions, not required S1 demo paths.
FR-18	S2	Automatic traffic management shall detect configured excess traffic, restrict eligible routine traffic and use a recovery waiting period to avoid repeated switching near a threshold.	Compare normal, faulty and bursty sources; record false restrictions and recovery.
FR-19	S1	The selected offline interface shall let a user submit a short help message and receive an operator reply through the shared dashboard without mobile-network or internet access.	Disable internet and exercise client-to-gateway-to-operator-to-client exchange.
FR-20	S1	The messaging channel shall distinguish local message storage, gateway acceptance and operator reply. Accepted messages shall not be silently deleted by rate control.	Delay or drop each stage and inspect displayed status and stored records.
FR-21	S1	Messaging admission shall use per-session and aggregate limits. A limited session shall retain a bounded opportunity to submit new help messages. The interface shall show when another attempt is eligible.	Exceed the session limit, wait for eligibility and submit again; test aggregate load.
FR-22	S1	If a Wi-Fi browser interface is selected, it shall require no installed application. Session controls shall not rely solely on an easily reset cookie; the service shall also bound aggregate admission.	Exercise browser access and reset sessions under a controlled load.
FR-23	S2	The selected feasible quality approach shall execute on the chosen MCU with the same estimator and documented preprocessing as the evaluated computer implementation.	Compare outputs on fixed inputs; measure memory, latency and energy.

For a radio-based messaging interface, equivalent client/session controls will be defined for its identity and transport model. A phone without a provisioned device key is not assumed to have wearable-level device authentication. Aggregate limits mitigate overload but do not guarantee resistance to every denial-of-service attack.

2.4 Nonfunctional requirements

ID	Category	Requirement	Evidence
NFR-01	Energy	Use USB for initial bench integration. Measure whole-board standby, sensing, processing, transmit and receive currents with a suitable measurement setup. Include independent listening and feedback. Estimate runtime for several report intervals using the selected cell and ageing assumptions.	Power traces and a reproducible duty-cycle budget.
NFR-02	Radio	Document selected band, output power, payload, airtime settings, retries and gateway downlinks against applicable BTK limits [11]. Local traffic controls shall not override regulatory limits.	Configuration review and airtime accounting.
NFR-03	Confidentiality	Encrypt and authenticate pulse payloads. AES-CCM is a candidate [10]. Protect clear routing/header fields through associated data where required. Nonces shall not repeat under a key; counters and restart recovery shall be specified.	Packet inspection, altered-field tests and restart tests.
NFR-04	Privacy	Require no personal identity or account for wearable reports. Limit gateway access and retained records to the defined study purpose. Do not claim unlinkability against the authorized gateway.	Configuration and retained-data review.
NFR-05	Reliability	Log message acceptance, delays, expiry and command results. Every displayed state shall agree with the latest confirmed evidence and the configured freshness/timeout rules.	Controlled loss, delay and timer tests.
NFR-06	Usability	Provide a button identifiable by touch and test glove access. Use no more than three documented vibration patterns in the first prototype. Comfort assessment shall include overnight-use considerations under the approved study scope.	Design inspection; approved usability protocol where applicable.
NFR-07	Safety	Before on-body use, inspect cell protection, wiring, exposed surfaces, heating and attachment. Pause participant tests if unsafe behaviour or discomfort occurs.	Safety checklist and applicable ethics approval.
NFR-08	Model evaluation	Use participant-separated outer testing. Fit feature scaling, feature selection, model settings and decision thresholds only within the development subjects.	Split audit, fixed estimator record and reproducible experiment logs.
NFR-09	Maintainability	Version the protocol, estimator, label definition, firmware and model. Publish submitted reports and backlog versions without credentials or private participant data.	Repository and build/documentation review.

Low-microamp standby and use without routine charging are product goals to evaluate, not demonstrated properties. Specific timing and current acceptance targets will be recorded with the chosen hardware in D4. Prototype board results will be distinguished from estimates for a later compact device.

2.5 Evaluation plan

First demonstration

Flow A: activate the wristband, obtain a pulse indication or an uncertain result, transmit a report, display its age, and return an operator acknowledgement by vibration. Apply a short routine-report pause; exercise manual help, retain independent listening, demonstrate the duplicate-command response after a lost acknowledgement, and resume routine reports at expiry. Use a wrong-tag command and an older-counter command to demonstrate rejection.

Flow B: send a short help message from the selected messaging client, display it beside the wearable on the same dashboard, and return the operator's reply. Apply a session limit and demonstrate a later permitted help submission. Both flows operate with internet access disabled. One wearable cannot establish network-scale performance; additional traffic sources can be explicitly labelled test generators.

Initial bench integration uses USB power. Battery selection for the portable demonstration follows current measurements and enclosure checks. A USB-powered demonstration does not establish battery runtime or final wearable size.

Communication and traffic experiments

Record delivery ratio, end-to-end delay, operator-confirmation delay, retry count and airtime. Compare the same offered load with traffic control enabled and disabled. Report lost or late help messages, routine-report coverage, false restrictions and recovery behaviour. A negative or neutral result remains part of the evaluation. Obstruction and range measurements will state materials, geometry, radio settings and limitations.

Pulse estimation and quality learning

Use PPG-DaLiA wrist PPG and acceleration with its ECG-derived reference [4, 5]. Align 8 s windows with a 2 s step: at 64 Hz there are 512 PPG samples; at 32 Hz there are 256 samples per acceleration axis. The initial estimator removes the mean, applies a Hann window and selects a PPG spectral peak in 0.5–4 Hz, with guarded interpolation using the peak and its neighbours in log amplitude. This is a fixed starting method, not motion compensation.

Usable labels describe whether that estimator meets a fixed reference-error tolerance; they do not represent universal physiological quality. Freeze the estimator, its numerical guards and the tolerance before outcome-based model tuning. ECG is used for labels and evaluation, not model input. Missing reference and failed estimates receive explicit handling.

Compare no quality screening, an acceleration-only threshold, the rule baseline, logistic regression and a small MLP. Use outer leave-one-subject-out evaluation with participant-separated internal tuning. Any training-only grey-zone exclusion is selected inside development data; boundary cases remain in the outer test. Report coverage, unacceptable-error fraction among accepted windows, pulse MAE, per-subject and activity results, and error–coverage curves. An empty accepted set has an undefined accepted-window error, not zero error.

The first-semester model runs on a computer. On-device deployment and measurements of the full sensing, feature, inference and radio pipeline are second-semester work. New sensor recordings require their own evaluation: results from PPG-DaLiA do not establish performance on the candidate MAX30101 or in an earthquake setting.

3 Literature review

3.1 Disaster and wearable communication

Pueyo-Centelles et al. describe LoRa-based citizen status messaging after an earthquake [6]. This motivates short predefined wearable reports and a separate channel for a detailed request. LifeLink will examine operator acknowledgements and the interaction between reporting frequency, receive opportunities and airtime. The paper is a related approach, not evidence that LoRa will reliably cross our target obstructions.

Soysal, Gürkan and Yavşan's IoT Band combines a wristband, GPS, temperature sensing and pulse oximetry, with Wi-Fi and an alternative GSM module [7]. Its stated use includes vital-data/location monitoring and supporting rescue intervention order. LifeLink instead targets a local communication path with explicit quality labels and no rescue-priority decision. This comparison concerns the reported architecture; Wi-Fi itself does not inherently require internet.

3.2 Candidate off-grid platforms

Platform	Documented capability	Question for LifeLink
Meshtastic [8]	LoRa mesh messaging and optional sensor telemetry. Official telemetry documentation includes health metrics such as heart rate and oxygen saturation; health support may require a suitable firmware build.	Can we add our quality labels, freshness policy, operator-issued acknowledgement and bounded routine-report control while meeting energy and airtime limits?
MeshCore [9]	A portable C++ packet-routing library and firmware examples for LoRa networks; documented roles and a configurable hop limit. Sensor, messaging and remote-management examples exist.	Which application/protocol changes are needed for the same LifeLink requirements, and what integration, receiving and traffic-control costs follow?
Minimal custom link	A possible direct wearable–gateway application protocol, designed only for the demonstrator's specified messages. No implementation result is claimed.	Does reduced scope simplify confirmation and scheduling enough to justify development, security review and interoperability work?

These are candidates, not an adopted stack. A routing or transport receipt is not by itself proof that a rescue operator read and acknowledged a request. LifeLink must define that application event for whichever platform is selected. The reviewed documents do not justify claiming that all competing systems lack signal-quality assessment or acknowledgements.

3.3 Contribution to evaluate

The proposed contribution is the integration and evaluation of quality-labelled pulse reporting, offline help messaging, confirmed operator feedback and traffic control that preserves bounded help access. We will assess their combined behaviour rather than claim novelty merely from combining LoRa, a wearable and machine learning.

3.4 Quality assessment and security

PPG-DaLiA contains wrist physiological/motion recordings from 15 participants, together with an ECG reference [4]. The associated Deep PPG paper studies heart-rate estimation [5]. We use these resources to evaluate a different question: when is a selected simple estimator reliable enough to accept? Daily activities provide movement variation, but not validation under compression, injury, cold or rubble.

Elgendi studies PPG quality indices including skewness and relative power [14]. These motivate small, inspectable features. Candidate LifeLink features are selected-peak power share, competing-peak ratio, motion variability, PPG–motion frequency separation and waveform skewness. Their usefulness must be tested; frequency proximity suggests motion contamination but does not prove its cause.

NIST SP 800-38C defines CCM for confidentiality and authentication [10]. It provides a candidate cryptographic basis for pulse reports and control messages. Application design must still specify keys, nonce uniqueness, counters, associated data, receiver state and restart behaviour. Encryption does not prevent channel jamming or eliminate congestion.

3.5 Implications for implementation

The reviewed work supports separate checks for communication, measurement error and human acknowledgement. Platform selection will therefore consider a bidirectional bench exchange, programmable application messages, independent receive scheduling, whole-board current, airtime and development effort. Hardware purchase will follow that comparison and the laboratory inventory.

The D4 analysis report will refine measurable thresholds, exceptional cases and message fields. The D6 high-level design will assign functions to components and interfaces. These later decisions must remain traceable to the requirements in this report. Changes will be recorded in the shared backlog rather than silently changing the scope.

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Web documentation checked for this revision: 22 October 2026. Dataset, software and regulatory versions will be recorded again when the implementation is selected.

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